

SyllabusOS

Product Motivation & Context

SyllabusOS v2 proved a thesis: students will adopt a syllabus-first planner if extraction is accurate, changes stay synced, and the product respects their trust. Five demo personas, multi-course aggregation, change-detection, and a paid tier for personal commitments shipped on schedule. Early activation looked strong — students who uploaded one syllabus uploaded the rest within a week.

v3 turns that single-term planner into a four-year academic operating system. The opportunity is bigger than one term: the same students who need a calendar in week one need a degree-progress view in junior year, a study coach when their grades slip, and a mobile reminder system that lives where they live. v3 closes three gaps that limited v2: the product still required users to forward emails when the data already lived in the LMS; planning ended at the term boundary; and workload estimates never learned from how long things actually took. Key tradeoffs in v3 are LLM cost vs depth of coaching, recency vs LMS API limits, and feature breadth vs trust on the institutional side.

Target Users & Problems

User Segments

- Undergraduate students managing multiple concurrent courses (carryover)
- First-year and international students unfamiliar with syllabus conventions (carryover)
- High-workload engineering and pre-medical students (carryover)
- Students balancing academics with work or athletics (carryover)
- Transfer students navigating credit equivalencies and partial-term enrollment
- Pre-grad-school students tracking GPA trajectory toward target programs

Core Pain Points (v3 expansion)

1. Email forwarding worked for v2, but the announcement data already lives in the LMS — students should not be the integration layer.
2. Multi-term planning is invisible. A student does not know in week 4 of sophomore year if a 'C' in this organic chem course closes the door on med school applications two years out.
3. Workload estimates are static. v2 used rule-based hours per task type. Reality varies massively by student, course, and week.
4. No social signal. Students cannot see whether their pacing matches peers or which study habits actually correlate with strong outcomes.
5. Mobile is an afterthought. v2's responsive web works, but students live in their phones — reminders need to behave like a real native app.
6. Burnout shows up late. Workload heatmaps surface academic load, but not the interaction between school, sleep, and other commitments.

User Journey Considerations

1. Connect an LMS, not a syllabus, as the first onboarding step.
 - OAuth into Canvas, Brightspace, or Blackboard.
 - Auto-import every active course, syllabus, assignment, and grade.

- Manual syllabus upload remains as a fallback for courses outside a supported LMS.
- 2. Mobile-first onboarding with native push from day one.
 - Lock-screen widget surfaces 'next task' without unlocking.
 - Daily 8am digest replaces v2's email digest as the default.
 - Offline mode caches the current and next week so a flight or commute does not break the planner.
- 3. Degree planner runs alongside the term planner.
 - Imports the university's degree audit (or accepts a parsed copy).
 - Shows progress toward graduation, required-but-unscheduled courses, and prerequisite chains.
 - Surfaces 'safe to take' versus 'overloaded' semester suggestions.
- 4. AI study coach as an ambient layer, never a chatbot.
 - Recommendations appear contextually next to the task they apply to (e.g. 'this is your highest-impact assignment this week').
 - Grounded in the user's actual grades, study log, and remaining weight — no free-floating advice.
 - Every suggestion shows its reasoning and a 'why this' source link.
- 5. Collaborative course view (opt-in, classmates only).
 - Shared workload heatmap when 3+ classmates have connected the same course.
 - Anonymous pacing percentile — 'you are tracking the median student on Lab 3'.
 - Peer-validated policy summaries — flagged when 10+ students edited the same low-confidence policy the same way.
- 6. Verified professor feed bypasses the change-detection review step.
 - Professors get a lightweight web tool to push deadline changes directly.
 - Verified pushes bypass user confirmation since the source is the authoritative party.
 - Email-forwarded changes remain user-confirmed as in v2.

Solution & Product Design

Foundation (carryover from v2)

v3 keeps the v2 foundation: syllabus extraction with confidence flags, workload heatmap, structured review with inline editing, change detection with user confirmation, multi-course aggregation, study plan, exam-period mode, Google Calendar export, mocked gradebook + translation, and the paid personal-events tier. v3 builds on top — none of these features are removed.

v3 Product (Week 6–9)

1. Real LMS integrations: Canvas, Brightspace, and Blackboard via official APIs. Auto-syncs every 4 hours; push-based when the LMS supports webhooks.
2. Native mobile apps (iOS + Android) with push notifications, lock-screen widgets, and offline mode.
3. AI Study Coach: personalized, grounded recommendations using an LLM with the user's grades, schedule, and study log as context.

4. Multi-term degree planner: imports degree audits, shows graduation progress, plans semesters ahead.
5. Adaptive workload estimation: learns from time-on-task signals (study-block completion, mobile-app time-in-task) to refine per-student estimates.
6. Collaborative course views: opt-in shared workload, peer pacing percentiles, peer-validated policy summaries.
7. Real-time professor feed: instructor-side micro-tool that pushes verified changes; bypasses confirmation for authoritative pushes.
8. Institutional dashboard: aggregate workload analytics for departments and student affairs offices (k-anonymous).
9. Wellness integrations: optional read-only Apple Health / Google Fit / Oura signals to model burnout risk.
10. Open API + integrations marketplace: third-party student tools (Notion, Anki, Beeminder) can read tasks and write completions.

Stretch (cut to v4 if not prototyped by mid-week 8)

1. Predictive grade modeling per course (Bayesian update on each gradebook pull) with calibrated confidence intervals.
2. Multi-language UI (full localization, not just policy translation).
3. Voice-first capture: 'Hey SyllabusOS, add a study block before my chem midterm.'
4. Faculty analytics: anonymized workload distribution per assignment so professors see how their course actually feels.

Design Considerations

- For all students (carryover principles, raised bar):
 - Trust still wins over coverage. AI suggestions show their reasoning; no silent automated changes to the calendar.
 - Mobile-native means real push, real widgets, real offline — not a responsive web shell.
 - Coaching grounds every recommendation in user data; the LLM cannot invent deadlines.
- For transfer and multi-term students:
 - Degree planner accepts partial-term enrollment and credit-equivalency edits.
 - Cross-term prerequisite warnings surface in onboarding, not the night before registration.
- For high-workload engineering and pre-medical students:
 - Predictive grade ranges show what the next-three-assignments outcome looks like.
 - Coach surfaces 'minimum effort to maintain target grade' suggestions when burnout signals appear.
- For student-athletes and working students:
 - Wellness signals (sleep, recovery, focus time) feed conflict detection — not just deadline overlap.
 - Travel-aware exam prep: prep blocks shift earlier when the planner sees a known travel weekend.
- For international students:

- AI coach defaults to plain English with mid-paragraph translation hover, not full-document translation.
- No training on user data without explicit consent — important for jurisdictions with stricter data-protection rules.
- For institutions:
 - Admin dashboards are k-anonymous ($k \geq 10$) by default. Individual student data never exposed.
 - Departments see aggregate workload by week, by assignment type, by course — useful for curriculum review.

Metrics & Success

Metric	Definition
LMS Connection Rate	% of users with ≥ 1 LMS account connected
Auto-Import Share	% of tasks added via LMS sync (vs manual upload)
Daily Active Users	Users who open the app each day (up from WAU in v2)
Mobile Session Share	% of total sessions originating from mobile apps
Coach Engagement	% of users interacting with ≥ 1 AI suggestion per week
Multi-Term Plans Created	Users with a planned semester beyond the current term
Workload Estimation MAE	Mean absolute error of estimated vs actual study hours
Grade-Prediction RMSE	Calibration of predicted course grade vs final grade
Institutional Pilot Count	Universities with an active SyllabusOS deployment
Wellness Opt-In Rate	% of users connecting at least one wellness signal

GTM, Risks & Summary

GTM

v2's distribution was organic — student networks, class groups, word-of-mouth. It worked, and v3 keeps that channel, but the unit of growth shifts from individual student to course and institution.

- Course-level virality: when 3+ classmates connect the same course, the collaborative view unlocks and pulls in more classmates.
- Institutional pilots: target 2–3 universities to offer SyllabusOS through student success offices. Pilot covers LMS integration, admin dashboard, and a co-marketing motion.
- Premium tier expansion: v2's paid tier (personal events + conflict alerts) is extended to include AI Coach access, multi-term planning, and wellness integrations. Free tier remains generous so the core trust thesis holds.
- Integrations marketplace: third-party tools become an organic acquisition channel — students discover SyllabusOS through Notion, Anki, or other apps they already use.

Risks (+ potential solutions)

- **LMS API friction:** Canvas, Brightspace, and Blackboard each have different APIs, rate limits, and institutional access policies. Some universities disable API access entirely. *Mitigation:* ship Canvas first (largest market share), keep email forwarding as a fallback, and build a clean adapter interface so a new LMS is a sprint, not a rewrite.
- **AI hallucinations on study advice:** an LLM that invents deadlines or misreads grading rules kills trust faster than a missed change. *Mitigation:* ground every recommendation in the user's actual data. Coach prompts are tightly scoped, retrieval-augmented, and never allowed to claim a fact not already in the database. Suggestions show their reasoning and a 'why this' source link.
- **LLM cost at scale:** per-user coaching is expensive. *Mitigation:* prompt-cache aggressively (Anthropic's prompt cache, 5-minute TTL is fine for coach sessions), batch background generations, and gate heavy features behind the paid tier.
- **Privacy with institutional data:** aggregated workload analytics must not leak individuals. *Mitigation:* k-anonymity ($k \geq 10$) on every admin query; differential privacy on aggregate exports.
- **Mobile dev complexity:** shipping two native apps doubles the test surface. *Mitigation:* React Native with a shared business logic core (the v2 lib/ layer ports directly); platform-specific code only for push, widgets, and OS integrations.
- **Workload model overfitting:** a learned estimator that mostly sees one student's data can swing wildly. *Mitigation:* blend per-user signal with course-level priors. New users see rule-based estimates until enough data accumulates.
- **Wellness signal sensitivity:** surfacing burnout risk is helpful but easy to do badly. *Mitigation:* opt-in only, plain-language framing ('you've had 4 short nights — consider moving the Tuesday block to Thursday'), no scores or stigma.
- **Time constraints:** v3 has more surface area than v2. *Mitigation:* same rule as v2 — any v3 feature not prototyped by mid-week 8 ships as a flag-gated beta to a subset of users rather than a full launch.

Summary

v3 stops treating SyllabusOS like a syllabus tool and starts treating it like the academic operating system students actually need: connected to their LMS, aware of their multi-term path, intelligent enough to coach a hard week, respectful enough to never silently auto-decide for them. v2 earned trust by doing one thing well. v3 expands the surface area without giving that trust back.

Technical Design

v3's architecture keeps v2's clear lib/adapter boundaries — every external system stays behind an interface that can be swapped without UI changes. The data layer migrates from localStorage to Prisma + Postgres so multi-device sync and institutional deployments become tractable.

1. LMS Sync Layer (replaces email forwarding as primary)

- Adapters for Canvas, Brightspace, Blackboard built on the same v2 interface contract.
- Pull every 4 hours; subscribe to webhooks where supported (Canvas LiveEvents).
- Diff incoming data against the local cache. Same change-detection precision rules apply — only flag real diffs.
- Email forwarding remains the fallback for unsupported LMS or courses without API access.

2. Mobile App

- React Native (Expo) so v2's lib/ business logic ports unchanged.
- Native modules only for push (FCM/APNS), lock-screen widgets, and OS calendar integration.
- Offline mode: SQLite cache of current + next week. Conflict resolution on reconnect.

3. AI Orchestration Layer (Coach)

- Claude API as the model. Use prompt caching aggressively — system + user-context prompts are cached for the 5-minute TTL.
- Retrieval-augmented: every coach response is grounded in the user's grades, tasks, and study-block log.
- Hard guardrails: the model cannot output a task, date, or grade not already in the database.
- Every response stores its grounding evidence so users can audit the suggestion.

4. Adaptive Workload Model

- Per-task signal: study-block completion, mobile-app time-in-task, self-reported actual hours.
- Per-user Bayesian update on top of course-level priors. New users default to the v2 rule-based estimator.
- Output: a calibrated estimated-hours value with an uncertainty interval, shown in the task drawer.

5. Multi-Term Degree Planner

- Imports degree audits (or accepts a structured CSV / manual entry).
- Tracks remaining requirements, prerequisite chains, and projected graduation date.
- Surfaces 'safe to take' versus 'overloaded' semester suggestions.
- Cross-term study plans support summer terms and partial enrollment.

6. Collaboration Layer

- Course-level opt-in. Joining a shared course view exposes only your pacing and policy edits — never grades or personal events.
- Anonymous pacing percentile computed server-side; client never sees other students' raw data.
- Peer-validated policy summaries surface when 10+ students edited the same low-confidence row in the same direction.

7. Verified Professor Feed

- Lightweight instructor web tool authenticated via institutional SSO.
- Professor-pushed changes are signed and bypass user confirmation (the source is authoritative).
- Email-forwarded changes still require user confirmation — the v2 precision-first behavior is unchanged for unverified sources.

8. Institutional Admin Layer

- Aggregate workload analytics: hours per week, distribution by course, identification of crunch-week patterns at the department level.
- All queries enforce k-anonymity ($k \geq 10$). Individual student records are never exposed.
- Differential-privacy noise on exports.

9. Persistence

- Migrate from v2's localStorage + Zustand persist to Prisma + Postgres.
- The repository pattern from v2 makes this mechanical — every type in lib/types.ts already maps 1:1 to a table.
- Auth via institutional SSO (SAML / OIDC) for the institutional tier, Google/Apple SSO for individuals.

10. System Flow Summary

- User connects LMS in onboarding → courses, tasks, and grades auto-import.
- LMS sync runs every 4 hours; webhooks push changes when supported.
- Detected changes follow v2 precision-first rules; verified professor pushes bypass confirmation.
- Study Coach reads the resulting state, generates grounded recommendations, surfaces them contextually.
- Mobile push, lock-screen widget, and email digest all read from the same task store.
- Institutional admin queries roll up workload signal with k-anonymity guarantees.

11. Design Tradeoffs

- Accuracy vs cost: every coach prompt costs money. Prompt caching plus a generous free-tier rate limit balances unit economics.
- Recency vs API limits: 4-hour sync is fast enough for most use cases; webhooks where available; manual refresh always available.
- Coverage vs trust: AI never auto-edits the calendar. Suggestions are always human-confirmed except for verified professor pushes.
- Institutional value vs student trust: admin analytics are aggregate-only with hard k-anonymity floors.
- Native breadth vs maintenance cost: React Native is the compromise — one codebase, two stores, with platform-specific hooks where they matter.